

Pandas Interview Questions

Section-A : Basic Interview Questions.

- Q1. What are Pandas?
- Q2. What are the Different Types of Data Structures in Pandas?
- Q3. List Key Features of Pandas.
- Q4. What is Series in Pandas?
- Q5. What are the Different Ways to Create a Series?
- Q6. How can we Create a Copy of the Series?
- Q7. What is a DataFrame in Pandas?
- Q8. What are the Different ways to Create a DataFrame in Pandas?
- Q9. How to Read Data into a DataFrame from a CSV file?
- Q10. How to access the first few rows of a dataframe?
- Q11. What is Reindexing in Pandas?
- Q12. How to Select a Single Column of a DataFrame?
- Q13. How to Rename a Column in a DataFrame?
- Q14. How to add an Index, Row, or Column to an Existing Dataframe?
- Q15. How to Delete an Index, Row, or Column from an Existing DataFrame?
- Q16. How to set the Index in a Panda dataframe?
- Q17. How to Reset the Index of a DataFrame?
- Q18. How to Find the Correlation Using Pandas?
- Q19. How to Iterate over Dataframe in Pandas?
- Q20. What are the Important Conditions to keep in mind before Iterating?

Pandas Interview Questions for Experienced

- Q21. What is Categorical Data and How it is represented in Pandas?
- Q22. How can a DataFrame be Converted to an Excel File?
- Q23. What is Multi-Indexing in Pandas?
- Q24. How to select Specific Data-types to Include or Exclude in the DataFrame?
- Q25. How to Convert a DataFrame into a Numpy Array?
- Q26. How to Split a DataFrame according to a Boolean Criterion?
- Q27. What is Time Series in Pandas?
- Q28. What is Time Delta in Pandas?
- Q29. What is Data Aggregation in Pandas?
- Q30. Difference between merge() and concat()
- Q31. Difference between map(), applymap(), and apply()
- Q32. Difference between pivot_table() and groupby()
- Q33. How can we use Pivot and Melt Data in Pandas?
- Q34. How to convert a String to Datetime in Pandas?
- Q35. What is the Significance of Pandas Described Command?
- Q36. How to Compute Mean, Median, Mode, Variance, Standard Deviation, and Various Quantile Ranges in Pandas?
- Q37. How to make Label Encoding using Pandas?
- Q38. How to make Onehot Encoding using Pandas?
- Q39. How to make a Boxplot using Pandas?
- Q40. How to make a Distribution Plot using Pandas?

Pandas Interview Questions for Data Scientists

Q41. How to Sort a Dataframe?

Q42. How to Check and Remove Duplicate Values in Pandas.

Q43. How to Create a New Column Based on Existing Columns?

Q44. How to Handle Missing Data in Pandas?

Q45. What is groupby() Function in Pandas?

Q46. What are loc and iloc methods in Pandas?

Q47. How to Merge Two DataFrames?

Q48. Difference between iloc() and loc()

Q49. Difference between join() and merge()

Q50. Difference between the interpolate() and fillna()

Section- B

1. What is pandas in Python?
2. How do you access the top 6 rows and last 7 rows of a pandas DataFrame?
3. Why doesn't DataFrame.shape have parenthesis?
4. What is the difference between Series and DataFrame?
5. What is an index in pandas?
6. What is Multi indexing in pandas?
7. Explain pandas Reindexing
8. What is the difference between loc and iloc?
9. Show two different ways to create a pandas DataFrame.
10. How do you get the count of all unique values of a categorical column in a DataFrame?
11. How Do you optimize the performance while working with large datasets in pandas?
12. What is the difference between Join and Merge methods in pandas?
13. What is Timedelta?
14. What is the difference between append and concat methods?
15. How do you read Excel files to CSV using pandas?
16. How do you sort a DataFrame based on columns?
17. Show two different ways to filter data
18. How do you aggregate data and apply some aggregation function like mean or sum on it?
19. How can you create a new column derived from existing columns?
20. How do you handle null or missing values in pandas?
21. Difference between fillna() and interpolate() methods
22. What is Resampling?
23. How do you perform one-hot encoding using pandas?

24. How do you create a line plot in pandas?

25. What is the pandas method to get the statistical summary of all the columns in a DataFrame?

26. What is Rolling mean?

Section-C

1. Explain Python Pandas?

2. What is the use of Python Pandas?

3. Define series in Pandas?

4. What are the types of data structures available in Pandas?

5. What are the critical features of Pandas?

6. How can the standard deviation be calculated from the Series?

7. Define DataFrames in Pandas?

8. What is the time series in Pandas?

9. Explain reindexing in Pandas?

10. Explain MultiIndexing in Pandas.

11. What is TimeDelta?

12. How to create a series from a dictionary in Pandas?
13. Which library tool is used to create a scatter plot matrix?
14. What is Time Offset?
15. Explain Categorical data in Pandas?
16. How to create a copy of the series in Pandas?
17. How to add an index in DataFrames?
18. How to delete an index from DataFrame?
19. How can I remove indices, rows, or columns from a Pandas DataFrame?
20. How to rename an index or column in a DataFrame?
21. How to add a column to DataFrame?
22. How to add rows to a DataFrame?
23. How to iterate over a Pandas DataFrame?
24. What is Pandas NumPy array?
25. How to convert DataFrame to NumPy array?
26. List some statistical functions in Python Pandas?

27. How can one identify the items in series A that are absent from series B?
28. How can we convert DataFrame to an excel file?
29. What is data aggregation?
30. What is Multiple Indexing?
31. What is concat() in Pandas?
32. What is GroupBy() in Pandas?
33. How to sort the DataFrame?
34. How can one create an empty DataFrame in Pandas?
35. How to split a DataFrame based on boolean criteria?
36. What do describe() percentiles values represent?
37. How can you merge data on common columns or indices?
38. How to write DataFrame to PostgreSQL table?
39. How to convert continuous values to discrete values in Pandas?
40. How are iloc() and loc() different?
41. Explain the difference between join() and merge() in Pandas?

42. Explain the difference(s) between `merge()` and `concat()` in Pandas?
43. Explain the difference between `interpolate()` and `fillna()` in Pandas?
44. What are the types of conversion methods in Pandas?

Section-D

1. What is *Pandas* in *Python* and why is it used for data analysis?
2. Explain the difference between a *Series* and a *DataFrame* in *Pandas*.
3. How can you read and write data from and to a *CSV file* in *Pandas*?
4. What are *Pandas indexes*, and how are they used?
5. How do you handle *missing data* in a *DataFrame*?
6. Discuss the use of `groupby` in *Pandas* and provide an example.
7. Explain the concept of *data alignment* and *broadcasting* in *Pandas*.
8. What is *data slicing* in *Pandas*, and how does it differ from *filtering*?
9. Describe how *joining* and *merging* data works in *Pandas*.
10. How do you *apply* a function to all elements in a *DataFrame* column?
11. Demonstrate how to handle *duplicate rows* in a *DataFrame*.
12. Describe how you would convert *categorical data* into *numeric format*.
13. How can you *pivot* data in a *DataFrame*?
14. Show how to apply *conditional logic* to columns using the `where()` method.

15. What is the purpose of the `apply()` function in *Pandas*?

Section – E

1. What is Pandas and what are its primary data structures?

2. How do you import Pandas in Python?
3. How do you create a DataFrame from a dictionary?
4. Explain how to read a CSV file into a Pandas DataFrame.
5. What is the method to display the first 5 rows of a DataFrame?
6. What's the function to check the data types of columns in a DataFrame?
7. How would you select a specific column from a DataFrame?
8. Which method would you use to filter rows based on a condition?
9. What methods are available to handle missing values in a DataFrame?
10. What function can you use to fill missing values with a specific value?
11. Which function helps in summarizing the basic statistics of a DataFrame?
12. How can you count the number of unique values in a column?
13. What are the steps to create a scatter plot using Pandas?
14. Explain how to group data by a specific column and calculate aggregate statistics.
15. What's the use of the **apply** function in Pandas?
16. What's the purpose of **groupby** in data analysis?
17. Describe how to perform hierarchical indexing.
18. What is vectorization? What are its advantages?
19. How do you save a DataFrame to an Excel file?
20. How do you read a JSON file into a Pandas DataFrame?
21. Explain how to read and write data to a HDF5 file.
22. Describe a scenario where you would use Pandas for data cleaning.
23. What are the steps to preprocess data for machine learning using Pandas?

24. Describe a sample use case for Pandas in financial data analysis.
25. What steps would you take to verify the integrity of your DataFrame?

Extra 40 Pandas interview questions you can use to assess applicants' skills

1. How do you remove duplicate rows in a DataFrame?
2. Explain how to convert a column to a different data type.
3. What's the method to rename columns in a DataFrame?
4. Describe how to apply a function to each element in a column.
5. How can you create a new column based on the values of another column?
6. What are the ways to handle outliers in a DataFrame?
7. What method would you use to find the correlation between columns in a DataFrame?
8. How do you generate a box plot for visualizing the distribution of data in a column?
9. How do you merge two DataFrames?
10. What's the difference between merge and concat functions in Pandas?
11. Explain the use of the melt function.
12. What's the purpose of the map function in Pandas?
13. How can you filter DataFrame rows based on multiple conditions?
14. Describe how to perform a left join on two DataFrames.
15. What are some use cases for the pivot_table function?
16. How do you use the cut function to bin continuous data into discrete intervals?
17. Explain how to perform a rolling window calculation.
18. How can you resample time-series data in Pandas?
19. What is the purpose of the qcut function?
20. How can you improve the performance of your Pandas code?
21. What is the use of eval and query functions in Pandas?
22. Explain how to work with large datasets that don't fit into memory.
23. How do you optimize memory usage in a DataFrame?
24. What is the impact of setting the appropriate data types on DataFrame performance?
25. What is the method to read data from a SQL database into a DataFrame?
26. Describe the process to save a DataFrame to a SQL database.
27. How can you integrate Pandas with Matplotlib for plotting?
28. What are the ways to export a DataFrame to a CSV file?
29. How can Pandas be used for time-series analysis?
30. Explain how to use Pandas for data aggregation and summarization.
31. How can you use Pandas to analyze web scraping data?
32. Explain how to handle text data in Pandas.
33. What are common errors encountered in Pandas and how do you resolve them?

34. How do you handle a situation where your DataFrame operations are slow?
35. Describe how to troubleshoot issues with missing or incorrect data in a DataFrame.
36. Discuss the importance of data types in Pandas.
37. What are best practices for handling large datasets in Pandas?
38. Explain the importance of indexing in Pandas.
39. What are the benefits of using chaining methods in Pandas?
40. How do you use the crosstab function for contingency tables?

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